Datamak

Live Datamak snapshot

EQUILIBRIA	GK INPUTS	REMOTE RUNS	SG ESTIMATES
4.21M	37.2k	359	174.9k

A database-centered workflow for gyrokinetic input generation, remote execution, monitoring, and surrogate modeling.

Ingest Equilibria

EFIT or TRANSP.
Parse remote folder.
Save metadata.

Create Inputs

Generation of GK input files with Pyrokinetic.
Optimized sampling of parameter space.

Deploy Batch

Auto restart, load balancing, and convergence of many runs..

Monitoring

Take care of remote runs.
Synchronize the results.
Diagnose failures or issues.

Surrogate

Train surrogate models. Estimate Physics for all inputs.

- Workflow and data can be monitored via web interface.
- Based on an SQL database.
- Tool implemented in python by AI agent.

Datamak Quickstart for the Demo Light Database

How to open the demo, read the main screens, and report errors

March 29, 2026

Goal

This quickstart is for a new user opening the `demo_light` database, browsing the main screens, and knowing how to report a failure clearly.

1. Start Datamak locally

1. Get Datamak with Git

- `git clone https://github.com/jdominsk/Datamak`
- `cd Datamak`

2. Set up Python

- `python3 -m venv .venv`
- `.venv/bin/pip install -r requirements.txt`

If Datamak is already cloned and `.venv` exists, reuse them.

3. Start the Flask server

- Run `.venv/bin/python gui/app.py`
- Leave that terminal open while you use Datamak.

4. Open it in Firefox

1. Start Firefox.
2. Go to `http://127.0.0.1:5000/`
3. If the page does not load, check the terminal for a Flask error.
4. If Firefox is stuck, clear recent history, restart Firefox, and reopen the plain URL.
5. If Datamak opens, move to **Database** only if you still need to load the demo database.

Beginner rule: First make the local server work, then load the demo database, and only then start exploring the screens.

2. Open the demo_light database

What Datamak does on first run

1. Launch Datamak.
2. If no valid database is configured, Datamak opens **Database** automatically.
3. It shows **Database recovery** and scans for local demo DBs.
4. Prefer **Use Detected Demo DB** when a suitable local copy is found.
5. Use the browser download only if nothing suitable is detected.

Key facts

- Expected file: the `demo_light` SQLite database
- Common locations checked:
  - `~/Downloads`
  - `~/Desktop`
  - Datamak root
  - `Datamak/demo`
- The Google Drive link may require a PPPL browser login.

Practical rule: If Datamak offers a detected local demo database, use that before trying to download another copy.

3. Know what each main screen is for

Fast orientation

Screen	What to do there
Physics Results	Inspect run outputs, statuses, and simple variable comparisons.
Surrogate Models	Review models, compute estimates, and read validation summaries.
Data Statistics	Explore the input space before or alongside simulation results.
Workflow	See the selected origin, current stage, and recommended next step.
Database	Inspect tables, schema details, and recovery options.

Suggested learning order

1. Start in **Workflow**.
2. Move to **Physics Results**.
3. Then inspect **Data Statistics**.
4. Review **Surrogate Models** after you understand the base data.
5. Use **Database** mainly for recovery or low-level checks.

4. Reading the science-facing screens

Physics Results

Good first questions:

- Which runs are `CONVERGED`, `RUNNING`, or `CRASHED`?
- How does `gamma_max` vary with a chosen input?

Start by choosing the data origin, then adjust the filter and the x/y axes.

Data Statistics

Use this screen to understand the *input space*:

- choose a data origin
- pick x and y input variables
- look at how cases cover the parameter space

This is often the fastest way to understand the demo database before reading detailed outputs.

Useful habit

Stay on one top tab while switching origins. Datamak now avoids heavy work for inactive tabs, so focused browsing is faster.

5. Workflow is the operator's home screen

Origin Registry

Select the origin you want to inspect. The selected origin drives the status, advisor text, and batch-monitor content shown on the page.

AI advisor

Read this first. It summarizes:

- current state
- main blocker
- recommended next step
- confidence

Batch monitoring

Use it to check simulations, understand status counts, and see whether the next action is generation, launch, sync, or troubleshooting.

Best habit

Before launching, syncing, or diagnosing anything, read the **Workflow** page for the selected origin. It is the safest place to understand what Datamak is already doing.

6. If something fails, send a useful error report

What the user now sees

A failure notice should be red, concise, and human-readable. The raw traceback stays hidden under **Technical details**.

Example summary

“Sync Back From Flux could not complete. Datamak could not log into the remote system. Check the saved username, SSH access, and Duo setup, then try again.”

What to do next

1. Read the plain-language summary.
2. If the fix is obvious, try it once.
3. Otherwise click **Send support bundle**.
4. Choose **Gmail** or **Mail**.
5. Send the generated report.

Support bundle contents: action context, stderr and traceback excerpts, a redacted configuration summary, recent usage-log context, and lightweight database counts.

7. Recommended first 10 minutes

Quick walkthrough

1. Start the local Flask server and open Datamak in Firefox.
2. Load the `demo_light` database.
3. In **Workflow**, select one origin and read the AI advisor.
4. In **Physics Results**, inspect one or two output plots.
5. In **Data Statistics**, switch x and y inputs and inspect coverage.
6. Read one surrogate-model validation summary.
7. If anything fails, use the red notice and send a support bundle.

Outcome

After this quickstart, a new user should know where to look for workflow status, results, statistics, model validation, and failure reporting.